Lesson 11: Variance

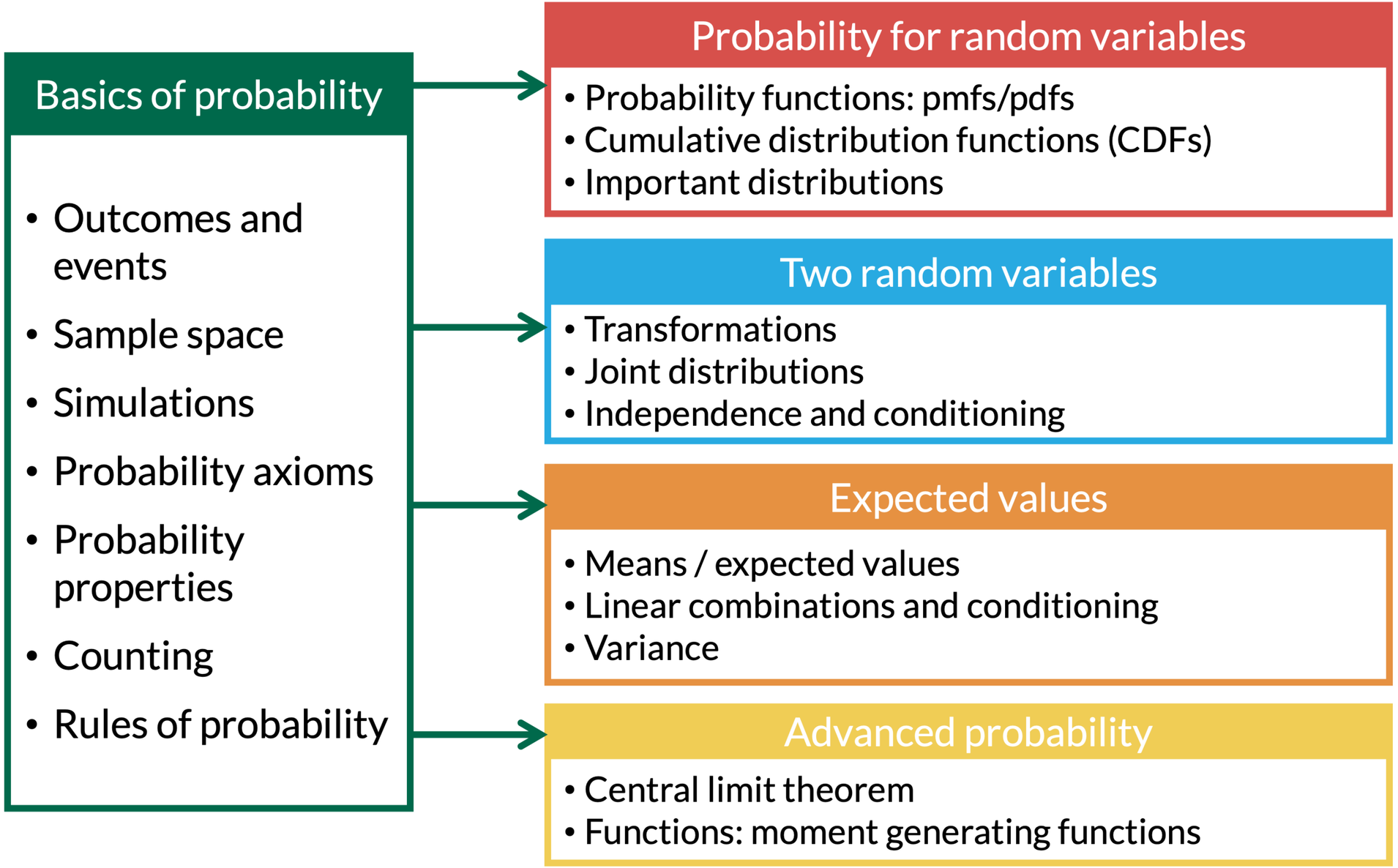
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Learning Objectives

1. Define and calculate the expected value for a function of discrete and continuous RVs
2. Define and calculate variance for a single random variable
3. Define and calculate variance for multiple random variables

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

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Let's start building the variance through expected values of functions

Example 1

Let g be a function and let $g(x) = ax + b$, for real-valued constants a and b. What is $\mathbb{E}[g(X)]$?

What is the expected value of a function?

Expected value of function of discrete RV

For any function g and discrete RV X, the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_{\{all\ x\}} g(x)p_X(x)$$

Expected value of function of continuous RV

For any function g and continuous RV X, the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- For example, if we have $g(x) = x^2$, then

$$\mathbb{E}[X^2] = \sum_{\{all\ x\}} x^2 p_X(x) \neq \left(\sum_{\{all\ x\}} x p_X(x) \right)^2 = (\mathbb{E}[X])^2$$

Let's revisit the card example (1/2)

Example 2

Suppose you draw 1 card from a standard deck of cards. Let $X = 1$ if the card is a heart and $X = 0$ if it is not.

1. Find $\mathbb{E}[X^2]$.

Recall Bernoulli RV (Lesson 10, Example 3):

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

$\mathbb{E}[X] = p$. Here, $p = \mathbb{P}(\text{heart}) = \frac{1}{4}$

Let's revisit the card example (2/2)

Example 2

Suppose you draw 1 card from a standard deck of cards. Let $X = 1$ if the card is a heart and $X = 0$ if it is not.

2. Find $\mathbb{E}\left[\left(X - \frac{1}{4}\right)^2\right]$.

Recall Bernoulli RV (Lesson 10, Example 3):

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

$\mathbb{E}[X] = p$. Here, $p = \mathbb{P}(\text{heart}) = \frac{1}{4}$

With $g(x) = \left(x - \frac{1}{4}\right)^2$ and $p = \frac{1}{4}$:

$$\begin{aligned} \mathbb{E}\left[\left(X - \tfrac{1}{4}\right)^2\right] &= \sum_{\{all\ x\}} \left(x - \tfrac{1}{4}\right)^2 p_X(x) \\ &= \left(0 - \tfrac{1}{4}\right)^2 (1-p) + \left(1 - \tfrac{1}{4}\right)^2 p \\ &= \tfrac{1}{16} \cdot \tfrac{3}{4} + \tfrac{9}{16} \cdot \tfrac{1}{4} = \tfrac{3}{64} + \tfrac{9}{64} = \tfrac{12}{64} = \tfrac{3}{16} \end{aligned}$$

Another way:

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Variance of a RV

Definition: Variance of RV

The **variance** of a RV X, with (finite) expected value $\mu_X = \mathbb{E}[X]$ is

$$\sigma_X^2 = Var(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Definition: Standard deviation of RV

The **standard deviation** of a RV X is

$$\sigma_X = SD(X) = \sqrt{\sigma_X^2} = \sqrt{Var(X)}.$$

Variance of discrete and continuous RVs

How do we calculate the variance of a discrete RV?

For **discrete RVs**:

$$\begin{aligned} Var(X) &= \sum_{\{all\ x\}} (x-\mu_X)^2 p_X(x) \\ &= \mathbb{E}[(X-\mu_X)^2] \\ &= \mathbb{E}[(X-\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

How do we calculate the variance of a continuous RV?

For **continuous RVs**:

$$\begin{aligned} Var(X) &= \int_{-\infty}^{\infty} (x-\mu_X)^2 f_X(x) dx \\ &= \mathbb{E}[(X-\mu_X)^2] \\ &= \mathbb{E}[(X-\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

Let's calculate the variance and prove it!

Lemma 6: "Computation formula" for Variance

The variance of a RV X, can be computed as

$$\begin{aligned}\sigma_X^2 &= Var(X) \\ &= \mathbb{E}[X^2] - \mu_X^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

Plan: start from the definition, expand the square, and use what we know about expected values

$$\begin{aligned}Var(X) &= \mathbb{E}[(X-\mu_X)^2] && \text{definition of variance} \\ &= \mathbb{E}[X^2 - 2\mu_X X + \mu_X^2] && \text{expand the square} \\ &= \mathbb{E}[X^2] - 2\mu_X\mathbb{E}[X] + \mu_X^2 && \text{split the sum; } \mu_X \text{ is constant} \\ &= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 && \mathbb{E}[X] = \mu_X \\ &= \mathbb{E}[X^2] - \mu_X^2 && \text{combine the } \mu_X^2 \text{ terms}\end{aligned}$$

Two facts from Lesson 10:

- $\mu_X = \mathbb{E}[X]$ is a constant
- $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

- This works for both discrete and continuous RVs!
- Want to see it written out with sums for a discrete RV? See the muddy points for this lesson.

Let's simulate the card example (1/2)

- Same steps as Lesson 2's dice, now with `n_cards <- 1`
- Each suit is a quarter of the deck, so each suit is equally likely

```
 1 set.seed(1028)                                                          ①
 2 reps <- 10000                                                           ②
 3 n_cards <- 1
 4 suits <- c("hearts", "diamonds", "clubs", "spades")                     ③
 5
 6 draws <- expand_grid(sim = 1:reps, card = 1:n_cards) |>                 ④
 7   mutate(suit = sample(suits, n(), replace = TRUE))
 8
 9 X_sims <- draws |>                                                      ⑤
10   group_by(sim) |>
11   summarise(X = sum(suit == "hearts"))
```

- The first 20 simulated values of X:

```
 1 head(X_sims$X, 20)                                                      ①
```

```
 [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0
```

- X can only be 0 or 1, just like a Bernoulli RV

Let's simulate the card example (2/2)

- Let's calculate some of the values we've shown by hand

```
1 X_sims |>                                                    ①
2   summarise(
3     mean_X        = mean(X),                                 ②
4     mean_X_sq     = mean(X^2),                               ③
5     variance      = mean((X - mean(X))^2),                   ④
6     var_shortcut = mean(X^2) - mean(X)^2,                    ⑤
7     std_dev       = sqrt(variance)                           ⑥
8   )
```

```
# A tibble: 1 × 5
  mean_X mean_X_sq variance var_shortcut std_dev
   <dbl>     <dbl>    <dbl>        <dbl>   <dbl>
1  0.249     0.249    0.187        0.187   0.432
```

- Both formulas for the variance give the same number!

Exact answers for the card example:

- $\mathbb{E}[X] = \frac{1}{4}$
- $\mathbb{E}[X^2] = \frac{1}{4} = 0.25$
- $Var(X) = \frac{1}{4} - \left(\frac{1}{4}\right)^2 = \frac{3}{16} = 0.1875$
- $SD(X) = \sqrt{3/16} \approx 0.433$

Variance of a Uniform distribution

Example 3

Let $f_X(x) = \frac{1}{b-a}$, for $a \leq x \leq b$. Find $Var[X]$.

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Variance of a function with a single RV

Lemma 7

For a RV X and constants a and b,

$$Var(aX + b) = a^2 Var(X).$$

Proof will be exercise in homework. It's fun! In a mathy kinda way.

Important results for *independent* RVs

Theorem 8

For independent RV's X and Y, and functions g and h,

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Corollary 1

For independent RV's X and Y,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Why Theorem 8 works (discrete example). Independence means $p_{X,Y}(x,y) = p_X(x)p_Y(y)$, just like $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$:

$$\begin{aligned}
\mathbb{E}[g(X)h(Y)] &= \sum_{\{all\ x\}} \sum_{\{all\ y\}} g(x)h(y)\, p_{X,Y}(x,y) && \text{expected value of a function} \\
&= \sum_{\{all\ x\}} \sum_{\{all\ y\}} g(x)h(y)\, p_X(x)p_Y(y) && \text{use independence} \\
&= \left(\sum_{\{all\ x\}} g(x)p_X(x) \right) \left(\sum_{\{all\ y\}} h(y)p_Y(y) \right) && \text{the } x \text{ and } y \text{ parts separate} \\
&= \mathbb{E}[g(X)]\, \mathbb{E}[h(Y)]
\end{aligned}$$

Variance of sum of independent RVs

Theorem 9: Variance of sum of independent RVs

For independent RVs (discrete or continuous) X_i and constants $a_i, i = 1, 2, \dots, n$,

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i).$$

Simpler version:

$$Var(a_1X + a_2Y) = Var(a_1X) + Var(a_2Y) = a_1^2 Var(X) + a_2^2 Var(Y)$$

Corollaries

Corollary 2

For independent RVs $X_i, i = 1, 2, \dots, n$,

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i).$$

Corollary 3

For independent identically distributed (i.i.d.) RVs $X_i, i = 1, 2, \dots, n$,

$$Var\left(\sum_{i=1}^{n} X_i\right) = nVar(X_1).$$

Ghost problem: *with replacement*

Example 5

The ghost is trick-or-treating at a different house now. In this case it is known that the bag of candy has 10 chocolates, 20 lollipops, and 30 laffy taffies. The ghost grabs a handful of five pieces of candy. What is the variance for the number of chocolates the ghost takes? Let's solve this for the case ***with*** replacement.

Recall probability for Bernoulli:

$$p_Y(y) = p^y(1-p)^{1-y} \text{ for } y = 0, 1$$

From Example 2: $\mathbb{E}[Y] = \mathbb{E}[Y^2] = p$

Back to our hotel example from Lesson 10

Example 6

A tour group is planning a visit to the city of Minneapolis and needs to book 30 hotel rooms. The average price of a room is $200 **with standard deviation $10**. In addition, there is a 10% tourism tax for each room. What is the **standard deviation** of the cost for the 30 hotel rooms? Assume rooms are independent.

Problem to do at home if we don't have enough time.

Find the mean and sd from word problem (1/3)

Example 7

A machine manufactures cubes with a side length that varies uniformly from 1 to 2 inches. Assume the sides of the base and height are equal. The cost to make a cube is 10 ¢ per cubic inch, and 5 ¢ for the general cost per cube. Find the mean and standard deviation of the cost to make 10 cubes.

Side work: $f_X(x) = 1$ for $1 \leq x \leq 2$

$$\mathbb{E}[X_i^3] = \int_1^2 x^3 \cdot 1 \, dx = \frac{x^4}{4}\bigg|_1^2 = \frac{16-1}{4} = \frac{15}{4}$$

$$\mathbb{E}[X_i^6] = \int_1^2 x^6 \cdot 1 \, dx = \frac{x^7}{7}\bigg|_1^2 = \frac{128-1}{7} = \frac{127}{7}$$

Find the mean and sd from word problem (2/3)

Find the mean and sd from word problem (3/3)

Let's simulate the cost of 10 cubes: simulate

- Same steps as the card simulation, with `runif()` from Lesson 8 for the Uniform(1, 2) side lengths

```
reps <- 10000                                                    ①
n_cubes <- 10

cubes <- expand_grid(                                            ②
  sim = 1:reps,
  cube = 1:n_cubes
  ) |>
  mutate(
    side = runif(n(), min = 1, max = 2),                         ③
    cost = 5 + 10 * side^3                                       ④
    )

cost_sims <- cubes |>                                            ⑤
  group_by(sim) |>
  summarise(total_cost = sum(cost))
```

```
cubes
```

```
# A tibble: 100,000 × 4
     sim  cube  side  cost
   <int> <int> <dbl> <dbl>
 1     1     1  1.47  36.5
 2     1     2  1.45  35.5
 3     1     3  1.10  18.4
 4     1     4  1.99  84.3
 5     1     5  1.88  71.4
 6     1     6  1.78  61.3
 7     1     7  1.09  17.9
 8     1     8  1.10  18.4
 9     1     9  1.41  33.2
10     1    10  1.67  52.0
# ℹ 99,990 more rows
```

Let's simulate the cost of 10 cubes: summarize

- Mean and SD of the total cost:

```
cost_sims |>
  summarise(
    average = mean(total_cost),
    std_dev = sd(total_cost)
  )
```

```
# A tibble: 1 × 2
  average std_dev
    <dbl>   <dbl>
1    425.    63.7
```

From Example 7: $\mathbb{E}[C] = 425¢$ and $SD(C) = \sqrt{4080.36} \approx 63.9¢$